



Inspiring Excellence

Department of Electrical and Electronic Engineering
Hardware / Software Report Submission

Semester: Summer 2026

Course Code: EEE101L

Course Title: Electrical Circuits I Laboratory

Section: 03

Experiment No: 8

Experiment Name: Verification of Thevenin's Theorem and Maximum Power Transfer Theorem and Study of Transient Behavior of RC Circuit using PSpice

Group No: 01

Group members:

<i>Sl.</i>	<i>Name</i>	<i>ID</i>
1.	Ruhinul Haque Bhuiyan	1000058247
2.	Manajir Tasbik Murshed	1000055271
3.	Dibyendu Roy Dibya	1000058422
4.		
5.		
6. (if applicable)		

Date of Submission: 11/08/2026

Prepared by: Ruhinul Haque Bhuiyan

LAB REPORT: EXPERIMENT 8

1. Verification of Thevenin's Theorem

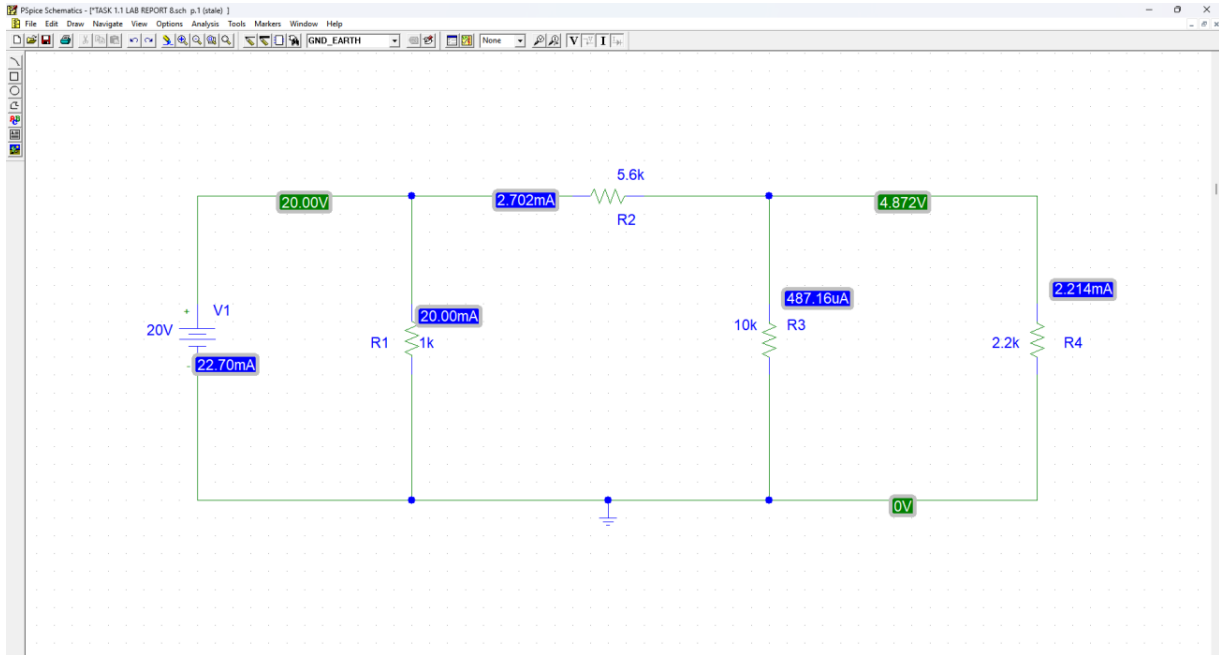


FIG 1.1: ORIGINAL CIRCUIT ($I_{L0}=2.214mA$)

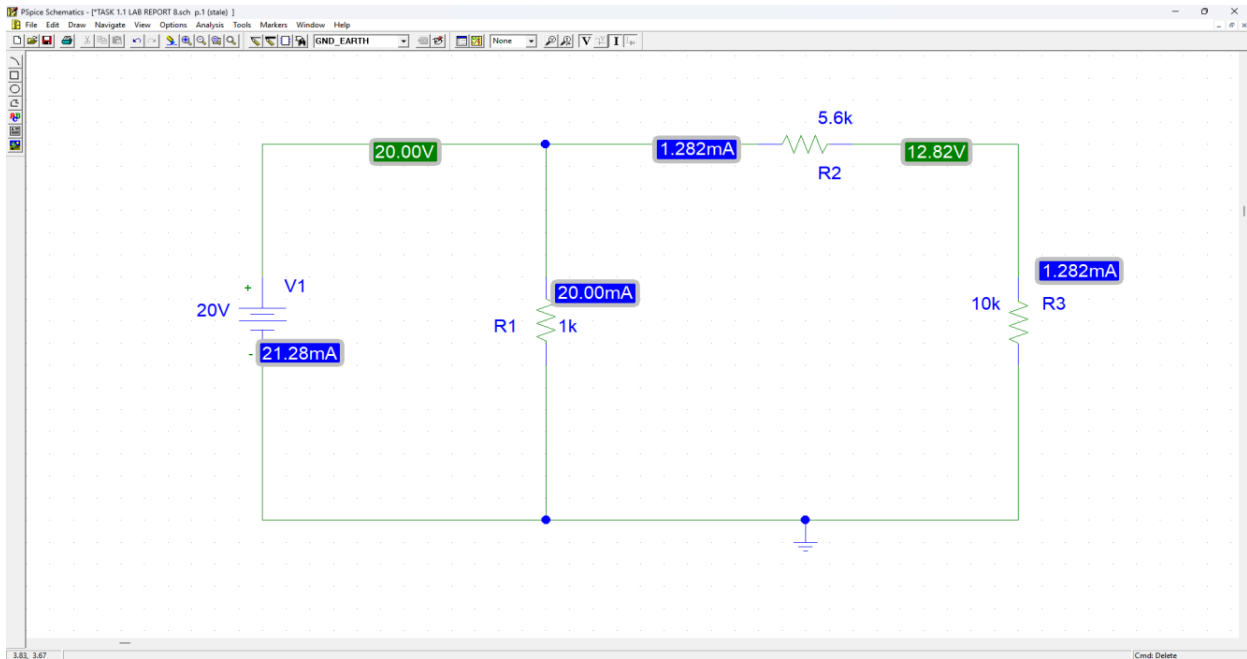


FIG 1.2: Finding $V_{TH} = 12.82 V$

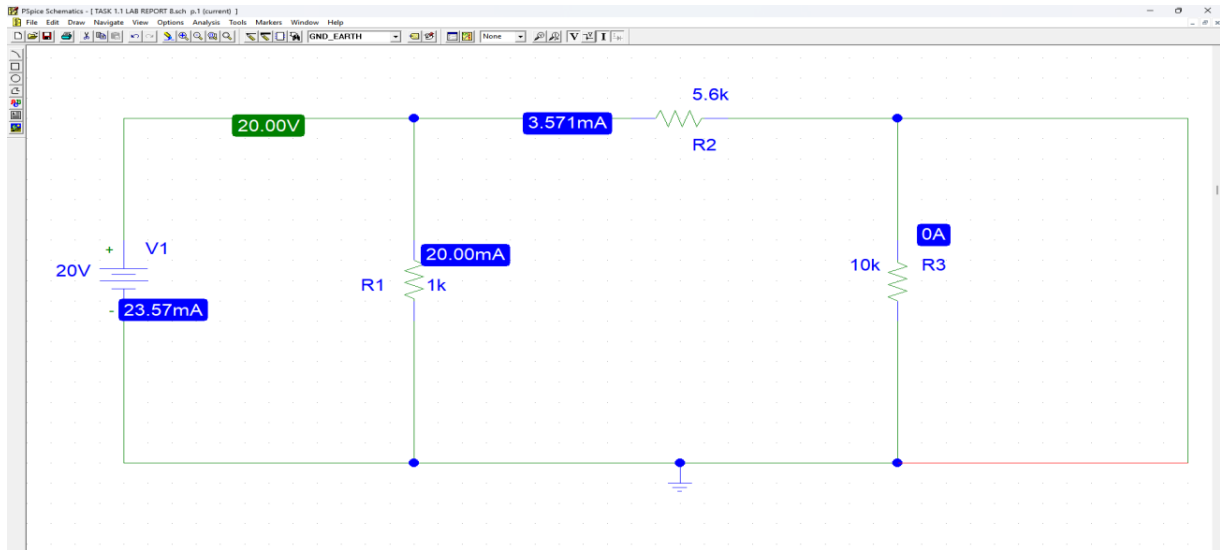


FIG 1.3: Finding $I_{SC} = 3.571mA$



FIG 1.4: THEVENIN EQUIVALENT CIRCUIT ($I_{LE} = 2.214mA$)

V_{TH}	I_{SC}	$R_{TH} = V_{TH} / I_{SC}$	I_{LO}	I_{LE}
12.82V	3.571mA	3590.0308 Ω	2.214mA	2.214mA

$$I_{LO} = I_{LE}$$

THEVENIN'S THEOREM IS VERIFIED.

2. Verification of Maximum Power

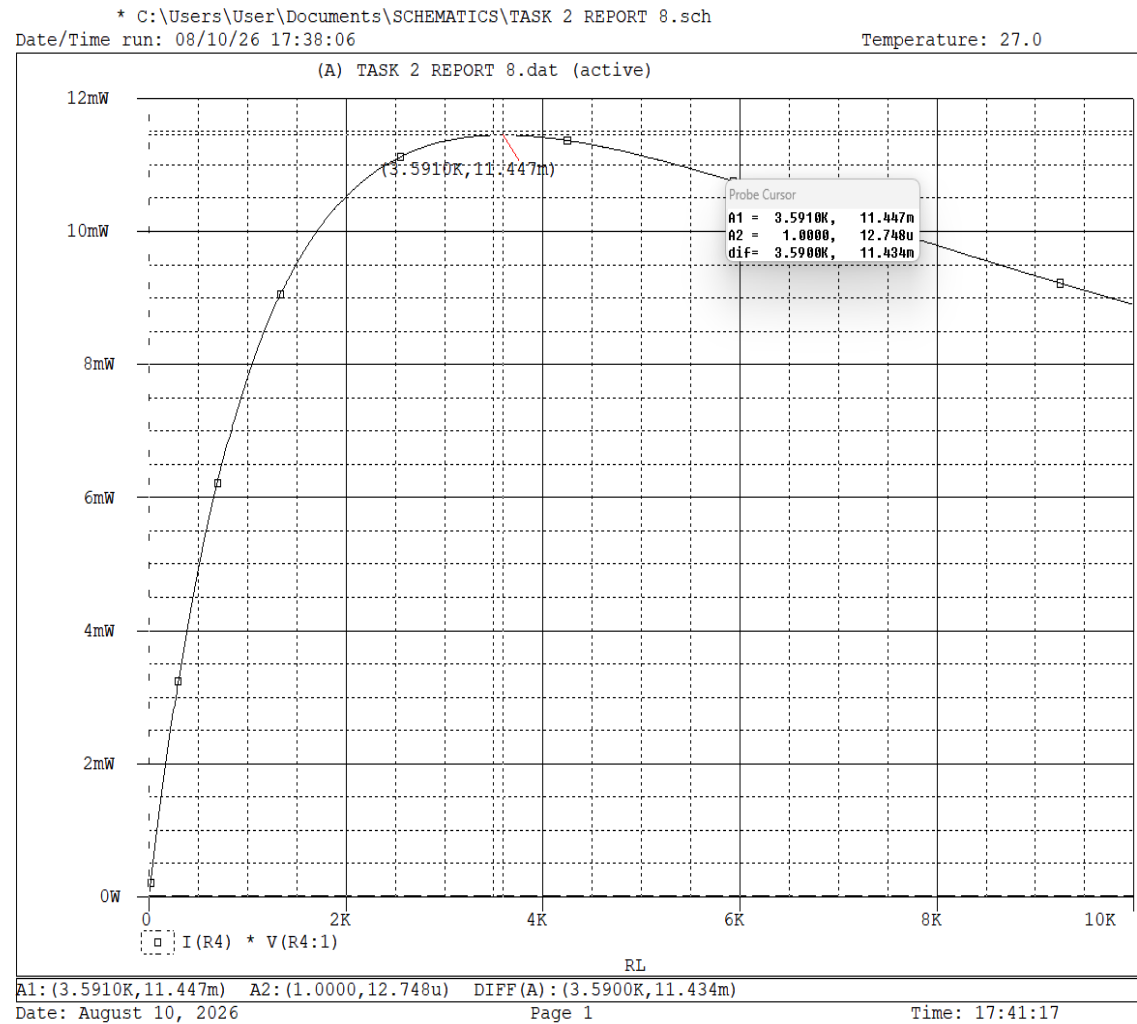


FIG: MAXIMUM POWER=11.447mW when $R_L = 3.5910K\Omega$

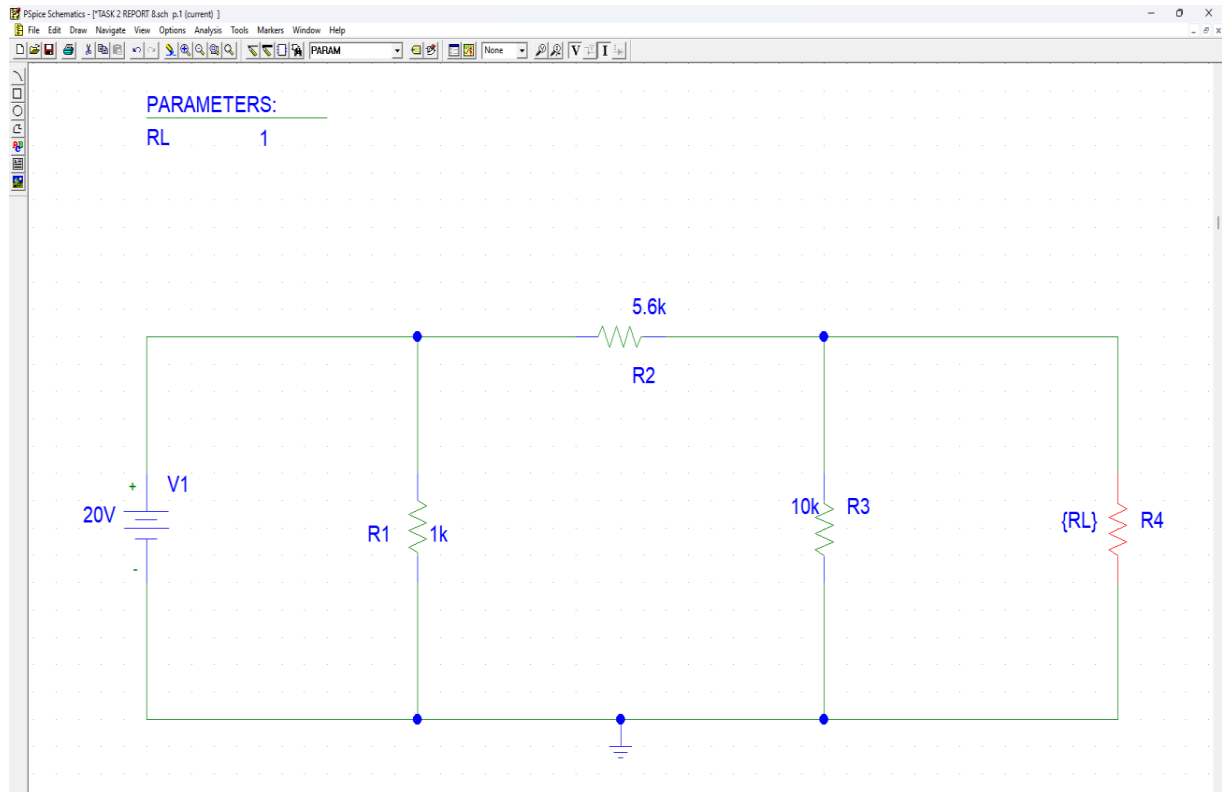


FIG: CIRCUIT DIAGRAM

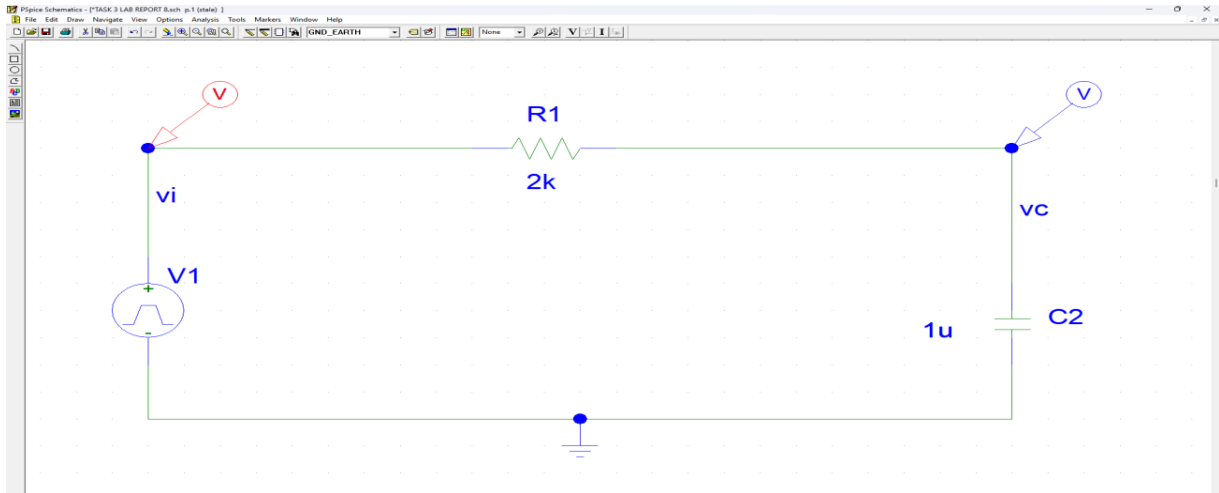
R _{TH} Measured	R _L For Maximum Power	Maximum Power
3.59003KΩ	3.5910KΩ	11.447mW

$$R_{TH}=R_L$$

The value of R_L for which the power delivered to the load is maximum is ***3.5910kΩ***. Also, the corresponding maximum load power is ***11.447 mW***.

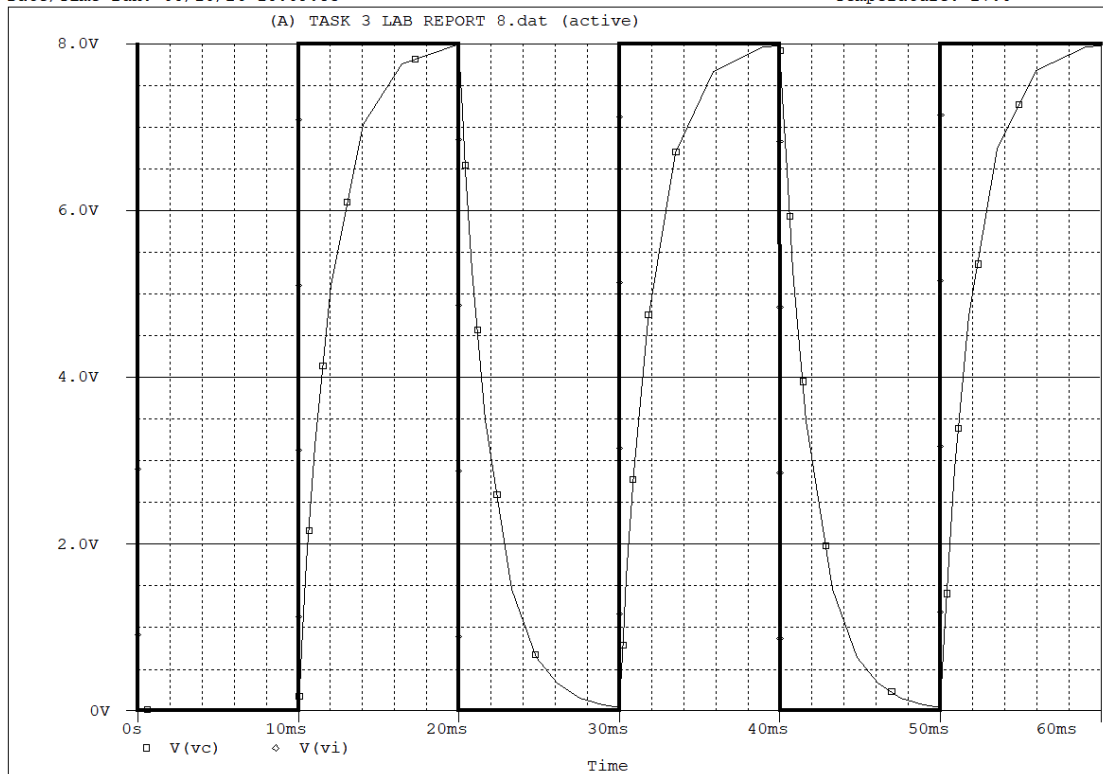
3. CAPACITOR-VOLTAGE WAVEFORM $V_c(t)$

CASE 1: $R = 2\text{ k}\Omega$ and $C = 1\mu\text{F}$



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Temperature: 27.0

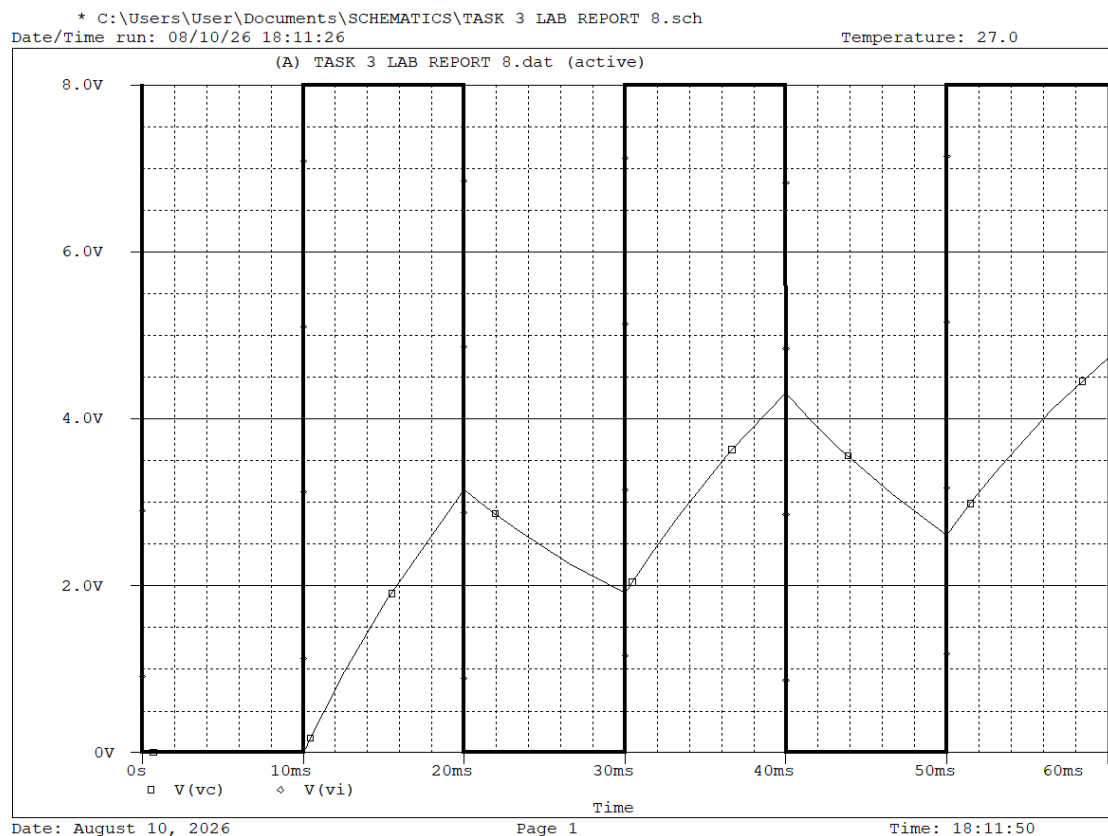
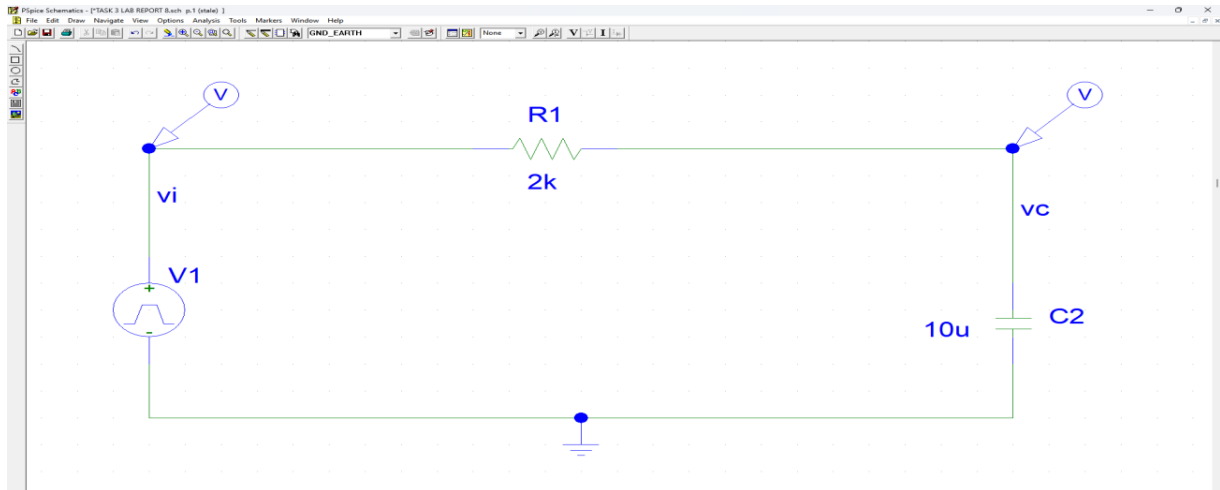


Date: August 10, 2026

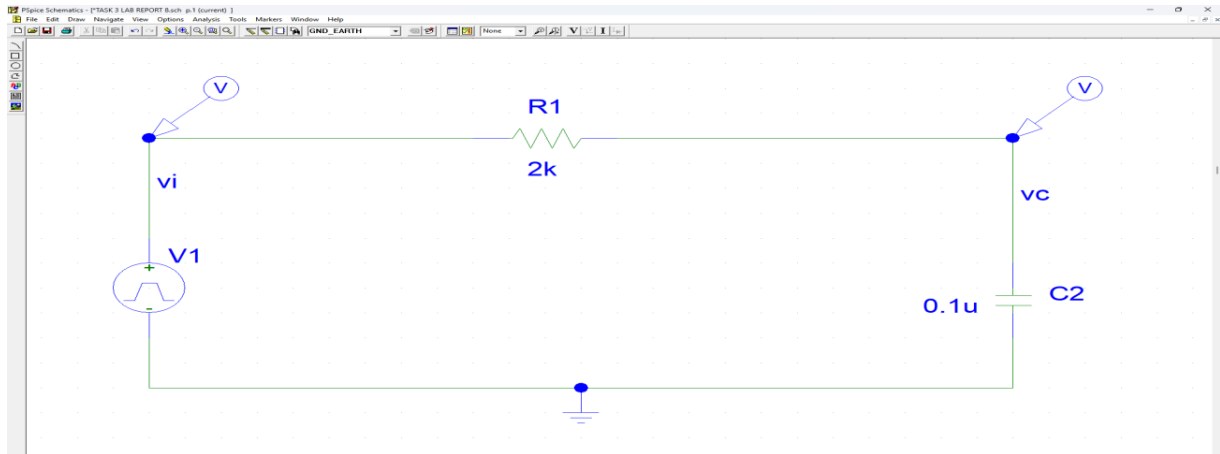
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CASE 2: $R = 2\text{ k}\Omega$ and $C = 10\mu\text{F}$

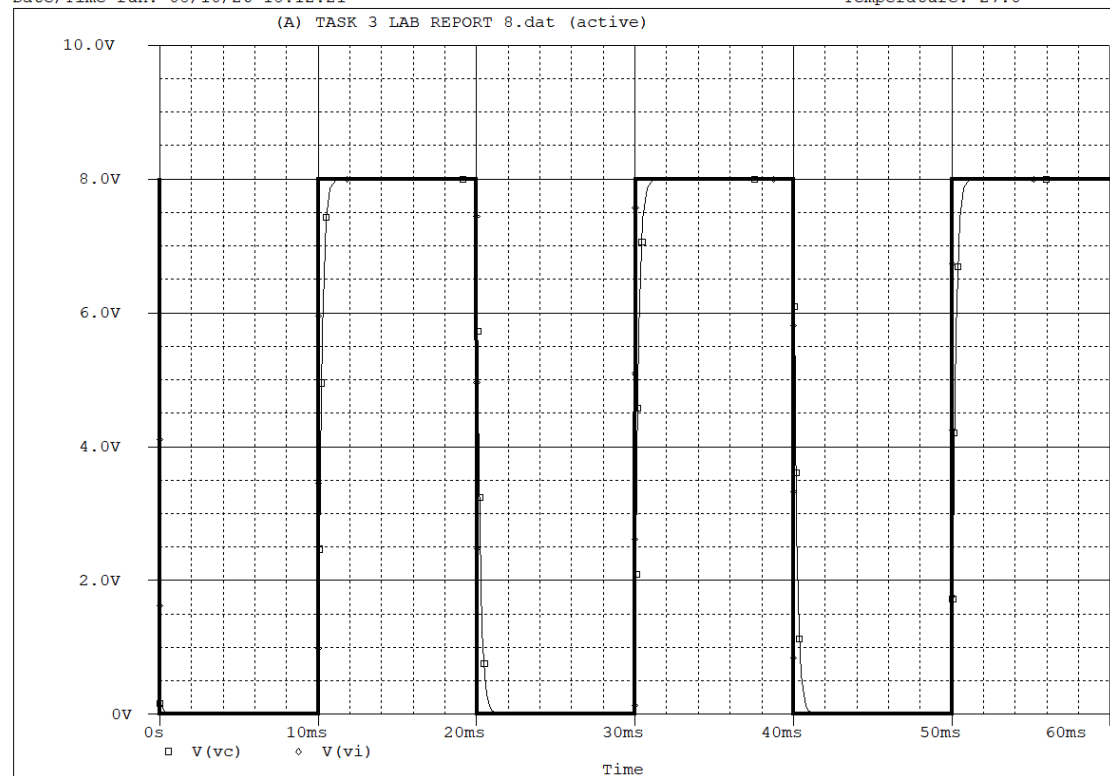


CASE 3: $R = 2\text{ k}\Omega$ and $C = 0.1\mu\text{F}$



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RC Time Constant (τ) Calculations:

For a pulse width $T_{pulse} = 10 \text{ ms}$ (Period $T = 20 \text{ ms}$, Square Wave $0 \text{ V} \rightarrow 8 \text{ V}$) and $R = 2 \text{ k}\Omega$:

- **Condition (i):** $C = 1 \mu\text{F}$

$$\tau_1 = R \cdot C = (2000 \Omega)(1 \times 10^{-6} \text{ F}) = 2.0 \text{ ms}$$

$$\text{Ratio } \frac{T_{pulse}}{\tau_1} = \frac{10 \text{ ms}}{2 \text{ ms}} = 5.0$$

- **Condition (ii):** $C = 10 \mu\text{F}$

$$\tau_2 = R \cdot C = (2000 \Omega)(10 \times 10^{-6} \text{ F}) = 20.0 \text{ ms}$$

$$\text{Ratio } \frac{T_{pulse}}{\tau_2} = \frac{10 \text{ ms}}{20 \text{ ms}} = 0.5$$

- **Condition (iii):** $C = 0.1 \mu\text{F}$

$$\tau_3 = R \cdot C = (2000 \Omega)(0.1 \times 10^{-6} \text{ F}) = 0.2 \text{ ms}$$

$$\text{Ratio } \frac{T_{pulse}}{\tau_3} = \frac{10 \text{ ms}}{0.2 \text{ ms}} = 50.0$$

Condition	Resistance (R)	Capacitance (C)	Time Constant (τ)	Charging Time (5τ)	vc(t) Peak Reached	Waveform Characteristics
Case i	2 k Ω	1 μF	2.0 ms	10.0 ms	8.0 V (100%)	Complete exponential charge & discharge per half-cycle
Case ii	2 k Ω	10 μF	20.0 ms	100.0 ms	$\approx 4.8 \text{ V}$	Incomplete charging
Case iii	2 k Ω	0.1 μF	0.2 ms	1.0 ms	8.0 V (100%)	Near-instantaneous charging; tracks square wave input

Explanation of RC Time Constant Effect: The time constant $\tau = RC$ determines the rate at which the capacitor charges and discharges. As capacitance increases, τ increases, requiring a longer total charging time (5τ). When $5\tau \leq T_{pulse}$, the capacitor reaches full steady-state voltage (8 V), whereas when $5\tau > T_{pulse}$, charging is interrupted before reaching full capacity, resulting in a triangular waveform.